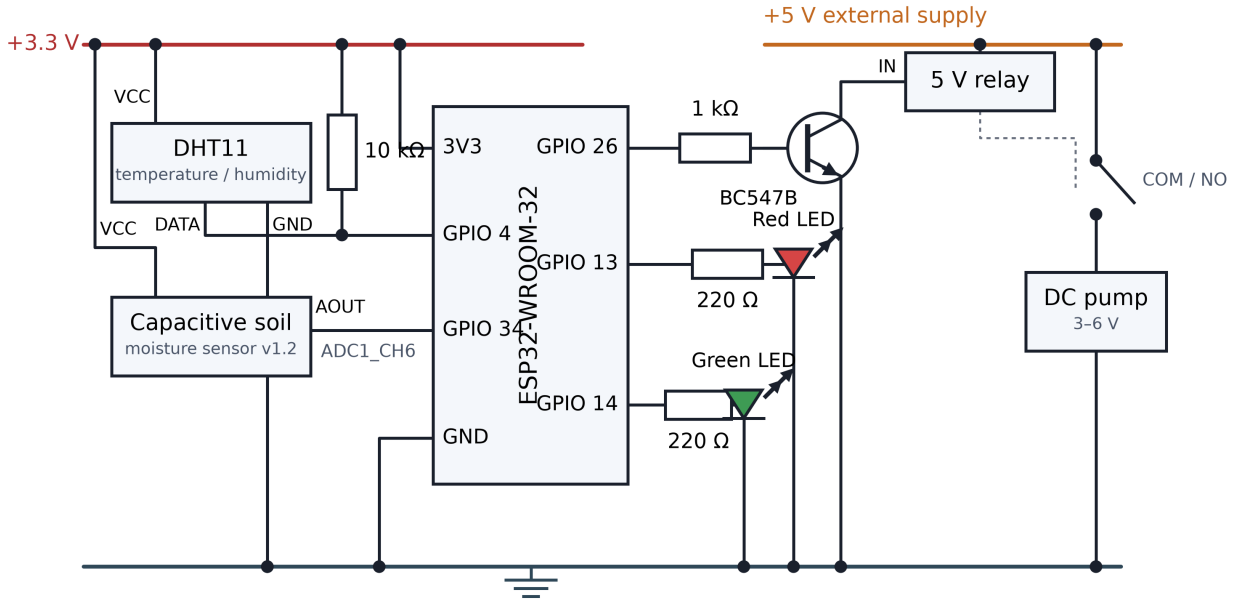
Smart Agriculture AI — Wiring Reference

ESP32-WROOM-32 irrigation node. Every pin below is taken from `hardware/smart_agri/smart_agri.ino`. If the firmware changes, this sheet must change with it.



Circuit schematic. Sensors on the 3.3 V rail; GPIO 26 drives the relay through a BC547B; the pump runs from a separate 5 V supply sharing a ground with the ESP32.

1. Pin map

ESP32 pin	Direction	Connects to	Through
GPIO 4	digital in	DHT11 DATA	10 kΩ pull-up to 3.3 V
GPIO 34	analogue in (ADC1_CH6)	Moisture sensor A0UT	direct
GPIO 26	digital out	BC547B **base**	1 kΩ
GPIO 13	digital out	Red LED anode	220 Ω
GPIO 14	digital out	Green LED anode	220 Ω
3.3 V	power out	DHT11 VCC, moisture VCC	—
GND	ground	common ground rail	—

GPIO 34 is **input-only** and **not 5 V tolerant**.

2. Parts

Part	Qty	Notes
ESP32-WROOM-32 dev board	1	powered from laptop USB
DHT11 temperature + humidity module	1	3-pin breakout
Capacitive soil moisture sensor v1.2	1	analogue output
5 V single-channel relay module	1	VCC / GND / IN • COM / NO / NC
DC mini pump, 3–6 V	1	runs from the power bank
BC547B NPN transistor	1	TO-92
Resistor 1 kΩ	1	transistor base
Resistor 220 Ω	2	one per LED

Part	Qty	Notes
Resistor 10 kΩ	1	DHT11 pull-up — only if the module has none on board
Red LED / Green LED	1 each	system alive / pump running
5 V power bank	1	relay coil and pump only
Breadboard, jumper wires, tubing	—	

3. Connection list

DHT11

`VCC`	ESP32 **3.3 V**
`DATA`	ESP32 **GPIO 4**
`GND`	ground rail
10 kΩ	between `DATA` and 3.3 V — **skip if your 3-pin module already has one**

Capacitive soil moisture sensor v1.2

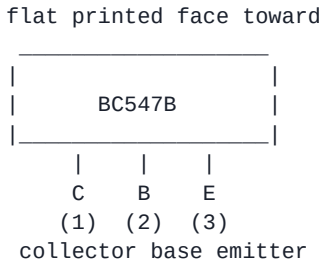
`VCC`	ESP32 **3.3 V** — 3.3 V, not 5 V
`AOUT`	ESP32 **GPIO 34**
`GND`	ground rail

Status LEDs

Red LED	GPIO 13 → 220 Ω → anode (long leg); cathode (flat edge) → ground rail
Green LED	GPIO 14 → 220 Ω → anode; cathode → ground rail

BC547B transistor

TO-92 package. Hold it with the flat printed face toward you and the legs pointing down:



Base (middle)	1 kΩ → ESP32 **GPIO 26**
Emitter (right)	ground rail
Collector (left)	relay module `IN`

Confirm the pinout against your actual part before powering up. TO-92 pinouts vary between manufacturers, and swapping collector and emitter is the usual way to destroy the transistor.

5 V relay module

`VCC`	power bank **+5 V**
`GND`	ground rail
`IN`	BC547B **collector**
`COM`	power bank **+5 V**
`NO`	pump **+**
`NC`	unused

Pump and power

Pump **+**	relay `NO`
Pump –	ground rail
ESP32	**laptop USB** — this is also the 115200-baud serial link
Power bank **+5 V**	relay `VCC` and relay `COM`
Power bank –	**ground rail** — mandatory, see section 5

4. Firmware constants

Measured on this build at 3.3 V. If the probe is re-wired or its supply voltage changes, re-measure the two calibration values.

Constant	Value	Meaning
`MOISTURE_AIR`	2590	raw ADC with the probe in dry air → 0 %
`MOISTURE_WATER`	1500	raw ADC with the probe in water → 100 %
`MOISTURE_DRY`	30	below 30 % → pump ON
`MOISTURE_WET`	60	above 60 % → pump OFF
Serial baud	115200	one JSON record per second
Loop period	1000 ms	
Green LED blink	250 ms	while the pump is running

Between 30 % and 60 % the relay holds its previous state. That is the hysteresis band, and it is deliberate.

5. Two things that will waste an afternoon

Common ground is mandatory.
The power bank's – must be tied to an ESP32 **GND** pin. Without it the transistor has no return path, the relay never switches, and the symptom looks exactly like a dead transistor or a bad GPIO.

Check the relay module's trigger polarity.
The firmware sets `GPIO 26 HIGH` to turn the pump **on**. The BC547B inverts that: GPIO HIGH → transistor conducts → `IN` pulled **LOW** → relay energises. That is correct for an **active-LOW** module, which most blue opto-isolated boards are. If yours is **active-HIGH** the behaviour comes out backwards and the pump runs when the soil is wet. Test it before putting the probe in soil (section 6, step 3). If it is active-HIGH, either drive `IN` straight from GPIO 26 and drop the transistor, or swap the HIGH/LOW on lines 86 and 91 of the firmware.

6. Bring-up order

Do not wire everything at once and hope.

Step	Do this	Expect
1	Sensors and LEDs only. No transistor, no relay, no pump. Flash and open the serial monitor at 115200.	One JSON line per second; red LED solid on.
2	Check the readings. Probe in air, then dipped in water.	Plausible temperature and humidity; air near **0 %**, water near **100 %**. Temperatures in the hundreds mean a loose DHT11 data line or a missing pull-up.
3	Add the transistor and relay. **Pump still disconnected.** Probe in air, then in water.	Air → relay clicks **on**, green LED blinks. Water → relay clicks **off**. Backwards? See section 5.
4	Connect the pump.	Full loop running.

7. Known issue

The logging session of 2026-10-24 recorded soil temperature of 717–742 °C and humidity of 1971–2074 % across 1,241 records — physically impossible. The most likely cause is a loose DHT11 data line or a missing pull-up. Step 2 above will catch it. Data before 2026-09-02 is clean and is what the project report uses.